

Machine learning and deep learning in river-basin modeling: A comprehensive review

Muhammad Waqas^{a,*}, Mohsin Nawaz^b

^a The Joint Graduate School of Energy and Environment (JGSEE), King Mongkut's University of Technology Thonburi (KMUTT), Bangkok 10140, Thailand

^b Faculty of Arts and Sciences, The Aga Khan University, Karachi 74800, Pakistan

HIGHLIGHTS

- This review of 133 studies assessing ML and DL models for river-basin hydrological modeling performance.
- ML/DL models perform well under non-stationary and heterogeneous hydroclimatic conditions across basins.
- Hybrid and physics-informed ML/DL outperform standalone and process-based models in accuracy and transferability.
- Emphasizes open data, explainable AI, and physics-guided models for sustainable river basin management.

ARTICLE INFO

Keywords:

Artificial Intelligence
River Basin Management
Physics-Informed Machine Learning
Hydrological Modeling

ABSTRACT

This study highlights a key question about the capacity of Machine Learning (ML) and Deep Learning (DL) models as standalone, futuristic approaches for river basin modeling. To answer the main research question, a PRISMA-based systematic review framework was used. This study evaluates 133 peer-reviewed studies that used standalone ML/DL models or compared them with conventional hydrological and process-based models to predict various hydrological variables and basin dynamics. Findings synthesis indicates that even though the ML/DL models are better in terms of pattern recognition, data integration, and predictive capacity, especially in non-stationary and non-heterogeneous hydroclimatic conditions, their application alone is not yet as physiologically interpretable and generalized across basins. Overall, the reviewed studies show that hybrid and physics-informed ML/DL models are more effective than traditional empirical models and data-driven models. Nevertheless, there are still difficulties, such as the lack of data, the inequality of computations in different regions, and the uneven assessment of extreme and low-flow states. This study highlights the necessity of hybrid modeling approaches, explainable and physics-guided AI, open data ecosystems, and standardized benchmarking frameworks to enable reliable, transparent, and fair river basin management in changing climatic conditions.

1. Introduction

Rivers are the foundation of the hydrological cycle and play a crucial role in human civilization, ecosystems, and sustainable development [1, 2]. Rivers are critical for promoting agriculture, providing drinking water, facilitating navigation, supporting biodiversity, and enabling renewable energy production [3]. Nevertheless, anthropogenic activities, land use land cover (LULC) changes, and global climate variability are increasingly threatening the operation of river systems and transforming the natural flow regimes and sediment processes across basins [4,5].

Classical hydrological and river models, based on physical and

empirical formulations, have been helpful in understanding processes of streamflow, sediment transport, and water quality [6,7]. However, they are limited in their ability to simulate the effects of complex, non-linear interactions in river basins, particularly when the environment is non-stationary with respect to climate and LULC [8,9]. This shortcoming has led to a swift shift toward data-driven, hybrid modeling systems based on artificial intelligence (AI), machine learning (ML), and deep learning (DL) [10].

This paradigm shift is necessitated by the fact that multi-source, multi-hydrological data, such as satellite data, in-situ sensors, and unmanned aerial systems, have grown exponentially in volume and heterogeneity [11,12]. Despite providing unprecedented temporal and

* Corresponding author.

E-mail address: muhammad.waqas@kmutt.ac.th (M. Waqas).

<https://doi.org/10.1016/j.asoc.2026.115295>

Received 10 November 2025; Received in revised form 30 March 2026; Accepted 18 April 2026

Available online 2 May 2026

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spatial resolution, they cannot be effectively modeled using classical methods [13,14]. High-dimensional data are fed into ML and DL, which identify patterns in noisy signals and Model non-linear relationships among climatic, hydrological, and geomorphological variables [15,16]. These approaches have revolutionized river modeling, encompassing flood forecasting and sediment yield prediction, as well as river morphology prediction and water-quality classification [17,18].

A variety of models and networks, including Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), Long Short-Term Memory (LSTM) networks, transformers, and ensemble models, such as Random Forests (RF) and Gradient Boosted Trees, have made significant progress in predictive accuracy, computational efficiency, and spatial generalization [10,19–21]. Initial applications of river models utilized empirical regression and support vector machines (SVMs) to predict discharge and sediment loads [22,23]. Deep neural networks (DNNs) have since enabled researchers to simulate the spatiotemporal relationships present within flow systems [5]. Standalone LSTM networks, in particular, can learn long-term relationships between rainfall and runoff, or among streams, whereas CNN-based models can learn patterns at a smaller scale (through images of river channels and floodplains) [4, 24]. More recently, hybrid and physics-informed ML (PIML) methods have integrated process understanding with data-driven flexibility, achieving better generalizability in settings with limited data [25,26].

Despite these achievements, the application of ML/DL to rivers remains in its infancy compared to other fields, such as meteorology or computer vision [27,28]. To accurately predict river behavior, therefore, requires not only large datasets but also physically meaningful representations of processes, including infiltration, evapotranspiration, channel migration, and sediment transport [5,8]. Traditional ML/DL models often assume that data are independent and identically distributed, which is inconsistent with the spatial and temporal dependencies of river networks [29].

Therefore, more recent studies advocate a hybrid approach to modeling, combining ML/DL with physically based models such as SWAT, HEC-HMS, and MODFLOW [30–32]. The exploitation of physical model interpretability and the adaptability of ML/DL in these hybridizations yields more powerful tools for predicting river discharge, flood risk, and sediment fluxes [4,33].

The increased availability of high-resolution topographic, climatic, and hydrological information has also promoted the application of ML in river science. Nowadays, datasets acquired via remote sensing (e.g., Sentinel, MODIS, and Landsat) and LiDAR-based digital elevation models enable large-scale mapping of river morphology and the definition of floodplain boundaries [34–36]. Transformer-based models and Generative Adversarial Networks (GANs) are already utilized to estimate missing hydrological data and generate synthetic data in data-sparse basins, thereby addressing one of the longest-standing challenges in river modeling [37–39]. There are also explainable AI (XAI) methods that aim to enhance model transparency and interpretability, which are essential for operational water management and policy adoption [13,22]. However, obstacles exist: the quality of heterogeneous data is not universal, algorithms are biased, computation is expensive, and not all can be widely adopted by hydrological practice [40,41].

The scientific necessity of a broad review stems from the disconnect in the existing literature, where applications of ML/DL to river systems tend to be case-specific and discipline-limited. In addition, novel methodological approaches, such as self-supervised, semi-supervised, and reinforcement learning, are rarely integrated into hydrological frameworks [4,42]. The main objectives are (i) to review the development of ML/DL models across hydrology and apply them to river modeling; (ii) to assess the role of ML/DL in the prediction of hydrological processes such as streamflow, sediment yield, water quality, and rainfall-runoff, and suggest future research directions based on hybrid, explainable, and transfer learning methods.

This review also discusses some guiding research questions: (1) How

have ML/DL methods developed in river studies compared to conventional hydrological models? (2) What are the most common algorithmic structures and data types applied to river applications? (3) How can hybrid and physics-informed ML/DL systems be used to understand hydrological processes? (4) What are the key technical and ethical issues that restrict ML/DL practical implementation in river basin management? (5) Which research directions might be taken in the future to facilitate explainable, transferable, and integrated AI systems? These questions are critical and urgent to answer, as the world's river systems are experiencing historically high levels of stress due to extreme climatic conditions and anthropogenic disturbances.

In comparison to the existing reviews (Table 1), which are either limited to specific tasks (e.g., inflow forecasting) [13], based on bibliometric reviews and absence of synthesis of processes [5], or concentrate on technological integration without a critical look at physical realism and ethics [43,44], the present review contributes to three different issues. (1) It provides a single synthesis of hydrological forecasting, sediment-geomorphological processes, and river water quality

Table 1

Comparative overview of past relevant review studies on ML/DL applications in water and river research.

Study	Scope & Focus	Strengths / Contributions	Limitations / Gaps
[43]	Broad review of ML in water resource management (irrigation, quality, floods, desalination).	Offers a taxonomy of ML algorithms, comparative mapping, and IoT integration.	Generalized focus; limited comparison with physics-based models; lacks discussion on explainability, transfer learning, and ethics.
[44]	Comparison of conventional hydrological models (SWAT, HEC-HMS, MIKE SHE, MODFLOW) with AI integration.	Highlights model-specific strengths and AI integration potential for sustainable management.	Emphasizes performance over process understanding; neglects morpho dynamics, sediment processes, and AI interpretability.
[4]	Systematic review of ML in river studies using text mining and content analysis.	River-focused; identifies key ML methods (NNs, DL, matrix factorization) and proposes an ML workflow.	Limited hybridization with physical models; minimal attention to explainability, transferability, and data ethics.
[13]	Review of ML/DL (LSTM, SVM, RF, BRT) for inflow and rainfall prediction.	Provides a clear comparison of ML vs DL; demonstrates LSTM's advantage for temporal data.	Narrow focus on inflow forecasting; excludes other river processes and hybrid model discussions.
[45]	Review of ML/DL models for groundwater salinity and seawater intrusion.	Highlights ensemble and optimized models (RF, XGBoost, CNN, LSTM), as well as variable importance analysis.	Groundwater-centric; limited relevance to river systems; omits sediment, morphology, and ethical issues.
[5]	Bibliometric analysis of ML/DL in hydrology (WoS, VOS viewer).	Maps global trends, leading countries, and dominant ML methods.	Lacks thematic depth and process insight; limited critical discussion on explainability, hybridization, and ethics.
[40]	Review of AI/ML in environmental health (air, water, disease, climate).	Addresses ethical concerns (bias, privacy, transparency) and human-AI collaboration.	Indirect relevance to hydrology; focused on the health-environment nexus.
[46]	Review of AI/DL with IoT for water distribution, wastewater, irrigation, and quality management.	Highlights AI's role in innovative water systems and sustainable operations.	Technology-focused; limited discussion of river dynamics, sediment, and hybrid modeling approaches.

at the river basin scale. (2) It critically evaluates the performance of models in terms of predictive accuracy, interpretability, transferability, and physical consistency, and explicitly uses comparative evidence across standalone, ensemble, hybrid, and physics-informed ML/DL systems. (3) It also includes methodological, operational, and ethical enhancements, such as explainable AI, data equity, and governance, that are not a key characteristic of earlier river-related reviews of ML/DL [4, 40].

2. Methodology

This review followed a systematic literature review (SLR) approach. To ensure rigor, transparency, and reproducibility throughout the entire review process, the methodological protocol was formulated in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines [47].

2.1. Research design and scope definition

The scope of this review was outlined to encompass studies that combine ML/DL models in areas such as river hydrology, hydraulics, river morphology, river sediment processes, and river water quality. Publications from 2010 to 2025 are included in this review.

This review aims at synthesizing the recent advances in the field of ML, DL, and AI applications to river system modeling and prediction with a specific focus on methodological trends, model performance, and new hybrid strategies.

2.2. Systematic search and data retrieval

Two major academic databases, Scopus and Web of Science, were used as the main systematic literature search. The reason why these databases were chosen is that they offer extensive coverage of peer-reviewed journals on hydrology, environmental sciences, and AI.

Fig. 2 represents records accessed in these two central databases. In addition to the primary search, to retrieve possibly useful studies that were not covered by the core databases, supplementary searches were conducted in Google Scholar and the chosen publisher databases (e.g., IEEE Xplore and MDPI). Combinations of keywords with ML/DL/AI and

river systems were used to formulate the Boolean search strings. Formulations of representative queries contained:

("machine learning" OR "deep learning" OR "artificial intelligence") AND ("river" OR "streamflow" OR "flood forecasting" OR "sediment transport" OR "water quality") AND ("hydrology" OR "catchment modeling" OR "basin" OR "watershed") (Fig. 1).

Peer-reviewed journal articles in the English language were selected to limit the search to articles with methodological credibility and domain relevance. The systematic search was finalized on 31 July 2025. Only studies published within the period 2010–2025 were considered to ensure that the review reflects recent advances in ML/DL-based river basin modeling.

2.3. Screening, eligibility, and quality assessment

A three-step filtering process, including identification, screening, and eligibility evaluation, was used as shown in Fig. 2.

In the identification phase, the duplicated records obtained in the two core databases were deleted before the screening was carried out to maintain the numerical integrity of the PRISMA flow diagram. Only those studies were shortlisted which include ML, DL, or hybrid AI applied to riverine or watershed processes: (1) Quantifiable results (e.g.,

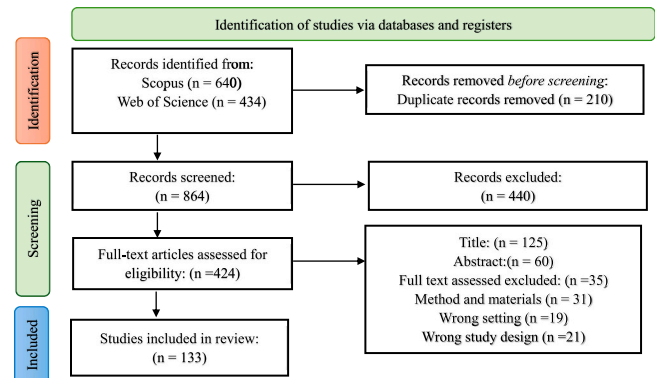


Fig. 2. Methodology for SLR for publication screening.

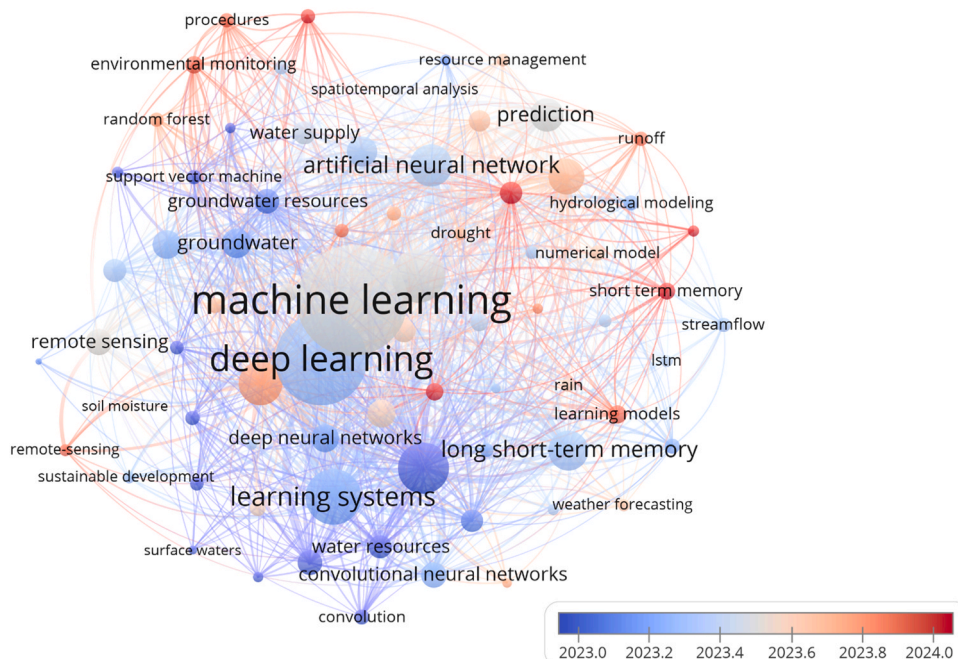


Fig. 1. Most reported keyword appearances in the literature on the application of ML/DL models to the research field since 2023.

accuracy of prediction, estimation of uncertainty, performance measures); (2) having sufficient methodological information to reproduce (type of model, dataset, preprocessing, evaluation); (3) described tendencies, constraints, or outlook in ML/DL/AI-based hydrological modeling.

At the full-text screening, the quality of the studies was evaluated based on the clarity of the model architecture, transparency of the information sources, suitability of the evaluation metrics, and sufficiency of the validation processes. The studies that did not provide a reproducible description of the methodology or included only in-sample evaluation were excluded to reduce the risk of performance inflation and reporting bias [20,47].

The number of exclusion counts included in the PRISMA diagram is directly proportional to each screening phase (duplicate elimination, title/abstract screening, and full-text eligibility assessment), so that the numbers are consistent numerically and transparent in the study selection procedure.

Even though the review is mainly focused on peer-reviewed journal articles, only a few preprints were retained when they concerned innovative approaches not yet available in the archives and when the methodology was transparent enough to facilitate criticism.

2.4. Evaluation and synthesis protocol

Standardized statistical indicators were used to normalize model performance across studies: Nash-Sutcliffe Efficiency (NSE), Kling-Gupta Efficiency (KGE), Root Mean Square Error (RMSE), Mean Absolute Error (MAE), etc. Thematic narrative analysis was employed as a qualitative synthesis method, focusing on convergence in methodology, trade-offs in interpretability, and physical constraints in AI models.

The synthesis also classifies the studies based on the type of model (ML, DL, hybrid, and physics-informed AI) and domains of use in the river systems, such as streamflow forecasting, flood prediction, modeling sediment transport, and water quality measurement.

2.5. Reproducibility and data transparency

In this study, no new empirical data were generated; it is a review of existing and peer-reviewed publications. All the search processes, inclusion criteria, and screening measures were recorded to provide transparency and reproducibility of the methods in line with PRISMA guidelines [47]. The approach focuses on transparency, reproducibility, and interdisciplinary accessibility, consistent with the open-science requirements of hydrology and AI studies.

3. Classification of hydrological and ML/DL models

3.1. Hydrological models

Hydrological models are key for simulating and predicting processes in river basins, including streamflow, groundwater recharge, sediment transport, snowmelt, and water quality [29,48]. These models are

classified into three primary categories: process-based, spatial representation-based, and randomness-based models (Fig. 3) [49].

3.1.1. Process-based classification

Process-based models are best suited for hydrological processes and are further divided into empirical/data-driven, conceptual, and physical models [50].

Empirical/data-driven models are based on the nature of observations/data without a direct physical representation [49]. In the past two decades, there has been rapid development of ML/DL models, including ANNs, RF, LSTMs, and gated recurrent units (GRUs) [51]. These models can handle non-linearity and temporal dependencies and often outperform conventional models for streamflow and sediment yield [52]. Nevertheless, it is still treated as a black-box model rather than a mechanistically interpretable one.

Conceptual models abbreviate the real hydrological processes into empirical relationships and lumped parameters [53]. For example, HBV and Topmodel, which offer computational efficiency and robustness in situations where long-term continuous datasets are limited [54]. Their smaller spatial realism, however, limits their applicability in the case of climate non-stationarity.

Physical models explicitly solve equations of mass, energy, and momentum conservation to model the hydrological cycle [55]. For example, SWAT, HEC-RAS, HEC-HMS, GSM-SOCONT, and MODFLOW, which are used to discretize watersheds to capture infiltration, evapotranspiration, snow/glacier melt, and runoff [56]. These models are interpretable and scalable, but they require intensive calibration, high-resolution meteorological data, and substantial computing resources [30]. It has been suggested that these limitations be addressed through hybridization with data-driven methods, such as GSM-SOCONT with DL in the Upper Indus Basin [56].

3.1.2. Spatial representation-based classification

Spatial representation-based models differ in their treatment of spatial variability, as distributed models explicitly capture heterogeneity in soils, LULC, and topography by discretizing watersheds into grid cells or hydrological response units [57], such as SWAP [58] and GSM-SOCONT [56]. In contrast, lumped models treat catchments as homogeneous and average watershed properties and processes into parameters. HBV and specific uses of HEC-HMS are in this category, which provides computational efficiency at the cost of spatial realism [54]. As the trade-off between detail and efficiency is a growing concern, semi-distributed approaches fill the gap.

3.1.3. Randomness-based classification

Another main class is based on the presence of randomness within models. Deterministic models rely on fixed input–output relationships, offering precise predictions under known conditions but exhibiting reduced reliability when input uncertainty increases [59]. Stochastic models, on the other hand, directly incorporate probabilistic aspects to capture the uncertainty in rainfall, parameter values, or the hydrological response [60]. These models are applicable within the framework of

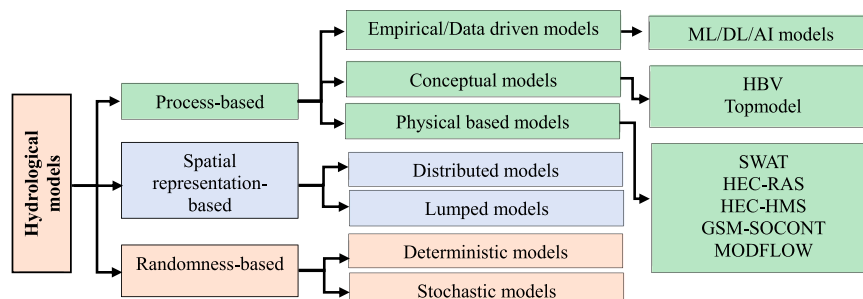


Fig. 3. Classification of hydrological models.

climate change, where non-stationarity and extremes are making deterministic assumptions break down [60]. The range of deterministic physical models to empirical and stochastic models shows advances in hydrological modeling research.

3.2. ML/DL models

Classification of ML/DL models in hydrology depends on the learning paradigm (supervised, unsupervised, semi-supervised, and reinforcement learning). The model architecture [10], as shown in Fig. 4. ML/DL models are applied to a broad range of hydrological processes, including rainfall-runoff modeling, river discharge forecasting, flood prediction, groundwater level monitoring, and water quality assessment [61,62].

Fig. 4(a) and (b) outline the key learning paradigms, the distinction between AI, ML, and DL, and further subdivide the methods into self-supervised, clustering, classification, and reinforcement learning. It further explains simple forms of learning, which include supervised learning (e.g., classification, regression) and unsupervised learning (e.g., clustering). Fig. 4(c) shows ML-based models, including tree-based ensembles (RF, XGBoost), SVM, and others. Finally, Fig. (d) shows DL-based architectures, such as ANNs, RNNs, LSTMs, CNNs, and PINNs, that can capture intricate spatio-temporal hydrological processes [10].

Conventional ML models in hydrology are mainly data-driven approaches trained to find empirical relationships between input and output hydrological variables [35]. These models are divided into supervised models (SVM, DT, RF, k-nearest neighbors (KNN), and ANNs) and unsupervised models (which group and categorize hydrological data without any labeled output) [63,64].

The most common type of supervised learning model in hydrological applications is one that predicts continuous (regression) and categorical (classification) outcomes [65]. DT and RF work best with non-linear interactions of different environmental predictors and are especially effective with heterogeneous hydrological data [66]. The generalization ability of support vector regression (SVR) has demonstrated high performance in modeling complex agricultural basins for total nitrogen and phosphorus concentrations [67].

K-means and hierarchical clustering can be used, as they are effective at identifying spatial and temporal groupings in rainfall patterns, groundwater behavior, or water quality indices [63]. These models enable hydrologists to uncover latent patterns in unlabeled hydrological data and to group river catchments or pollution areas together based on multivariate similarities.

Hybrid ML models gained popularity for leveraging the matching strengths of multiple algorithms. Considering the example of complete ensemble empirical mode decomposition with adaptive noise (CEEM-DAN) and extreme gradient boosting (XGBoost), combining both techniques improved the predictive performance of six leading water quality indicators [68]. These models overcome the issues of overfitting and model instability associated with single-algorithm modeling.

3.2.1. DL-based models

DL models use hierarchical feature learning via multi-layered neural networks (MLPNNs). A significant contribution to the development of hydrological models is the use of CNNs, RNNs, LSTM networks, and their hybrids (e.g., CNN-LSTM) [69–71].

CNNs are efficient for spatial data analysis, remote sensing, satellite image analysis, and river mapping [72]. CNNs extract spatially dependent features, when used in conjunction with environmental information, have also been able to automatically detect the quality of river water and floods [62]. RNN-based models, i.e., GRU and LSTM, excel at temporal sequence modeling and at predicting time-dependent hydrological parameters, including river flow, rainfall-runoff, and sediment transport [71]. Similarly, Long-term dependencies in sequential hydrological data have been recognized by LSTMs, which are particularly useful for river discharge forecasting and multi-step time-series prediction [71].

However, individual CNNs and LSTMs can be computationally intensive and less interpretable. To address these constraints, more modern architectures, such as Bi-LSTM and CNN-LSTM systems, have been proposed, which can be scaled to large hydrological datasets [70]. DL is better than traditional ML for automatically extracting features from large volumes of hydrological data collected by sensors, satellites, and other IoT devices [73]. However, the most common pitfalls of DL models are over- and under-fitting, lack of transparency, and over-parameterization, which are addressed by explainable AI and physics-informed neural networks (PINNs) that leverage knowledge of hydrological processes within data-driven architectures [74].

In river-basin modeling, ML/DL depends on the available input data, problem complexity, and interpretability of the results [71]. SVM, RF, and DT are still used in medium-scale regular data hydrological modeling [75]. One potential way to obtain accurate, scalable, and interpretable predictions in hydrology is through ensemble or hybrid paradigms [61]. This hierarchy of ML/DL models in hydrology reflects another evident line of evolution from rule-based and statistical models

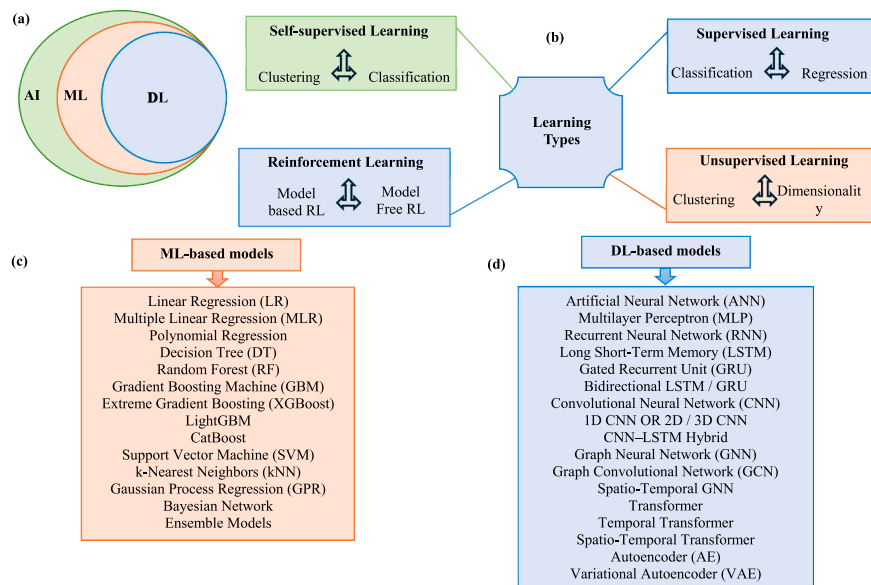


Fig. 4. Classifications of ML/DL based models in hydrology [10]ML-based models.

to hybrid data-driven models that can capture the complexity of how rivers function. Furthermore, Table 3 provides a comparative summary of ML/DL models used in river and water-related research.

3.2.2. Hydrological process perception behind the superiority of ML/DL models

Non-linear storage-release processes, threshold responses, spatial heterogeneity, and long-memory processes govern river basins. The rainfall is not directly converted into discharge; instead, it is filtered by soil moisture storage, groundwater exchange, evapotranspiration, channel routing, and sediment entrainment. These changes are highly non-linear and change with space and predecessor conditions [6,76].

Conventional hydrological theories have required explaining these changes through simplified parameterizations of the mass and momentum conservation laws. Nevertheless, infiltration capacity, hydraulic conductivity, and sediment detachment coefficients are challenging to quantify and are commonly set to fixed values, yet they change with land use, climate variability, and geomorphic changes in reality [8,9].

ML/DL models are better predictors, in that their mathematical forms are implicitly designed to capture the following hydrological phenomena: ML models based on trees (RF, XGBoost) divide the predictor space into regions defined by thresholds; these threshold-based regions are similar to runoff generation thresholds, including field capacity over flow and saturation overflow [66]. LSTM/GRU networks are proposed to be analogous to catchment memory via gated cells that store and release information, akin to soil moisture and groundwater reservoirs [77]. CNN models are used to isolate spatially distributed patterns in DEMs, LULC, and satellite data, and are operationally equivalent to distributed hydrological models that divide watersheds into hydrological response units [72]. Transformers and attention mechanisms discover long-range dependencies in hydrological factors, which are needed for climate non-stationarity when past patterns are not repeated directly [38].

Therefore, ML/DL models are not only successful because of the power of computing, but also because of their structural consistency with the inherent nature of river basin dynamics. They model the non-simplified, unknown hydrological transfer functions that are otherwise simplified in physical models [27].

3.2.3. Highest priority of hybrid DL models

Hybrid DLs are better at maintaining consistency in bias-variance trade-offs than physical and standalone models. The physical models exhibit significant structural bias and slight variance. They use the laws of conservation and reduce the actual processes to fixed parameters and calibration assumptions [55]. Standalone DL models have high variance and low bias. They are trained on complex data patterns but do not fully follow mass balance, produce physically inconsistent outputs, and fail to extrapolate [29].

However, hybrid DL solves both problems at once. Notably, a physical model is employed in conjunction with the DL models to provide a first-order physically consistent approximation, and the DL learns the structure of unmodeled processes, parameter uncertainty, and data noise [25,30]. Furthermore, PINNs are also trained by including conservation equations in the loss function, and a constraint is imposed on the learning process that confines it to the space of hydrologically viable solutions [26].

That is why hybrid DL has a two-fold limitation: (1) it interpolates further in ungauged basins, and (2) it works effectively under the climate and non-stationarity of LULC, requires less effort in calibration, and produces physically interpretable results. Therefore, hybrid DL is not only the integration of methods but also a statistical-physical co-learning system.

3.2.4. DL architectures can be synergized hydrologically to formulate DL strategies

All DL architectures represent different hydrological features. River

basin prediction requires aligning the architecture choice with the process representation to optimize prediction. As shown in Table 2, some efficient architectures are presented along with their hydrological characteristics and functional reasoning.

Therefore, proper implementation of the DL in river basins should be guided by the following logic: CNN/GNN for spatial representation, LSTM/GRU/Transformer for temporal dynamics, GAN/AE for data augmentation, and PINN/Physical model coupling for physical realism. Reviewed studies have empirically demonstrated that combinations such as CNN-LSTM, GNN-LSTM, and Transformer-LSTM are consistently superior to single architectures, as they capture multiple dimensions of hydrological processes simultaneously [61,69,78,80].

4. Results and discussion

4.1. Comparison of ML/DL methods and conventional hydrological models in river studies

Applications of ML/DL in river-based modeling have evolved from black-box, data-driven models and purely predictive approaches to hybrid approaches [74,87]. It also includes aspects of physics-aware approaches that systematically address the shortcomings of traditional hydrological models [54]. Several studies employed and developed ML/DL methods; however, only a few compared hybrid or hydrological models.

Therefore, these studies are grouped as follows: First, standalone DL (e.g., LSTM, GRU, CNN, DNN) has the capacity to catch complicated and non-stationary signals in streamflow, sediment load, and water quality proxies and frequently attains superior statistical metrics in-sample or in the hindcast experiments [52,88,89]. Also, ML/DL models are good in the field of pattern recognition, data gap filling, and local predictions (with rich historical observations or remote sensing proxies) [90,91], but are sensitive to the representativeness of the training samples, as well as the quality of the future forcing, in forecast/operational contexts [54].

Second, hybrid approaches that retain the physically meaningful structure but use ML/DL to learn the residuals [92], to fuse balancing data sources [93]: GRU-GSWAT+, and CNN-LSTM hybrids are better than either the pure process model or the pure data model to predict peaks and low flows and decrease systematic bias [30,56].

Third, ensemble and multi-stage designs (stacking, boosting, permutation feature selection, wavelet decomposition) encourage generalizability and pragmatic defenses against overfitting but require careful cross-validation and quantification of uncertainty to ensure that an analyst is not deceived by high training scores [51,84].

These studies highlight some persistent gaps as shown in Table 4: forecast performance is still dependent on future forcing (rainfall/temperature predictions), interpretation and attribution of learnt patterns

Table 2
Selection of DL architecture for hydrological modeling based on main system characteristics and learning abilities.

Hydrological characteristic	Suitable DL architecture	Functional motive
Spatial heterogeneity of terrain, LULC, and river morphology	CNN, GNN	Extracts spatial features and network connectivity [72,78].
Catchment memory and delayed runoff response	LSTM, GRU	Stores antecedent conditions like soil and groundwater storage [77].
Long-range temporal dependency under climate variability	Transformer	Attention mechanism captures non-local temporal relations [38].
Sparse or missing observations	GAN, Autoencoders	Synthetic data generation and feature compression [37,79].
Physical consistency	PINN, SWAT/DL coupling	Enforces conservation laws during learning [26].

Table 3

Comparative summary of advanced AI and ML models applied in hydrology, water quality, and resource forecasting.

ML/DL Model(s)	Study Area/ Country	Objective	Key Findings	Limitations	Reference
ANNs	Potomac River, USA	Prediction of water quality parameters (DO, BOD, COD, turbidity); adsorption and filtration modeling	High predictive accuracy ($R^2 > 0.90$), robust for non-linear datasets; widely used in water quality and adsorption modeling	Overfitting with small datasets; sensitivity to input parameter selection.	[64]
SVR, RF, XGBR, and RNN	Upper Ganga Basin, India;	Prediction of water quality index	RNNs showed high correlation and stability, and outperformed other models (RMSE of 0.75, MAE of 0.57, MSE of 0.30, and an R^2 score of 0.96)	Reduced interpretability; sensitive to hyperparameter selection	[75]
DT, RF, and XGBoost	The Tualatin River in Oregon, USA.	Prediction of short-term water quality prediction	CEEMDAN-RF and CEEMDAN-XGBoost models achieve the best overall prediction accuracy and stability for water quality parameters, with average MAPEs of 3.90% and 3.71%, respectively.	Computationally expensive; limited generalization to unseen conditions	[81]
KNN and K-means clustering	India	Classification of water types and LULC effects	Simple and effective for small datasets; performs well for local hydrological clustering.	Performance declines with large, high-dimensional data	[63]
Deep Reinforcement Learning (DDPG, TD3, SAC18, SAC19)	Folsom Reservoir, California, USA	To identify optimal reservoir operation policy and improve reservoir release decisions under complex, high-dimensional conditions	TD3 and SAC methods showed promising performance, efficiently learning optimal policies despite the curse of dimensionality	Traditional DP and SDP are ineffective due to the curse of dimensionality, implying high computational complexity.	[82]
SVM, and RF	Tamil Nadu, India	To evaluate and compare the accuracy of SVM and RF algorithms for LULC classification using Landsat data	RF achieved a higher accuracy (0.92) than SVM (0.81). Significant LULC changes from 1993 to 2023 include decreases in water bodies and agriculture, and increases in built-up areas, mining, and barren land.	Further testing of RF in diverse morphoclimatic conditions is needed	[67]
RTSGAN + GNN + LSTM	Danube and Tisza Rivers, Serbia	Enhance Dissolved Oxygen (DO) prediction using synthetic data generated from limited real-world measurements and water flow data.	GNNs improved WT and DO prediction accuracy by up to 43.4%, LSTMs by up to 15%, with consistent gains across 1–3-day horizons.	Dependent on the quality of sparse DO data and GAN-generated synthetic data	[78]
TNN, ML, DL, Empirical formulas	Malaysia	Develop highly accurate AI models for forecasting evaporation using meteorological data.	AI models, especially TNN, outperformed empirical and conventional ML/DL models, with TNN showing the highest precision across daily, weekly, and monthly forecasts.	No universally accepted optimal method; empirical methods are limited by recalibration needs and non-linear complexity	[83]
ANN, LSTM, WANN, WLSTM	Teesta River Basin, India	Discharge forecasting at three hydrological stations for different lead times	WANN and WLSTM models outperform others, are effective for long-term discharge prediction and flood mitigation	Input data only from the monsoon season; study limited to three stations	[84]
CNN-LSTM with Attention	Beilun Estuary, China	Predict water quality variables (pH and $\text{NH}_3\text{-N}$)	Outperforms other models, stable prediction with different time lags	Not explicitly mentioned; implies need for further research on non-linear problems and noisy data	[70]
TB, DTF, SDT, MLPNN	Jhelum River Basin, Pakistan	Predict streamflow	DTF outperformed other AI models with $R^2=0.99$, $\text{NSE}=0.99$, $\text{RMSE}=382 \text{ m}^3/\text{sec}$; decreasing trends observed in MK and Sen's slope tests	Limited to historical data (1971–2012); seasonal variations may affect model accuracy.	[65]
ANN, KNN, XGBoost, RF, CNN, CNN-RNN	Bardha watershed, India	Simulate monthly rainfall-runoff	CNN-RNN performed best with training $R^2=0.99$, $\text{NSE}=0.99$, $\text{MAE}=1.76$, $\text{RMSE}=3.11$, $\text{PBIAS}=-1.45$, testing $R^2=0.97$, $\text{NSE}=0.97$, $\text{MAE}=2.05$, $\text{RMSE}=3.60$, $\text{PBIAS}=-3.94$	–	[69]
SVM, GEP, MLP, WNN	Kaleybar-Chai River, Iran	Predict monthly streamflow	MLP outperformed others with $R^2=0.99$ (train), 0.98 (test); $\text{RMSE}=0.13$ (train), 0.21 (test); $\text{NSE}=0.99$ (train), 0.98 (test)	–	[85]
PINNs + Model Predictive Control (MPC)	Not explicitly mentioned	Develop a surrogate model for river water level and flow to optimize flood control strategies.	Accurate water level and flow prediction; effective real-time flood mitigation; reduced computational cost; successful dam control strategy for downstream flood management	Study limited to case study; generalizability to other regions not assessed; potential dependency on data quality.	[74]
Forward Dictionary Learning Detector (FDLdet)	Public large-scale RS datasets: SVCD, SYSU-CD, LEVIR-CD	Detect land-cover changes from remote sensing images by incorporating essential information (element composition and land-cover category)	FDLdet outperforms existing deep-based methods by generating dictionary features with crucial information, improving change detection accuracy	Rely on the quality of feature extraction; computational complexity is not discussed.	[86]
Hybrid SVR-ANN	Salvador Metropolitan Region, Brazil	Water demand forecasting	The proposed model is feasible and outperforms MLP, SVR, LSTM, and ARIMA.	–	[87]

Table 4

Qualitative comparison of ML/DL models with hydrological models, highlighting study regions or hydrological focuses, modeling approaches, key findings, and critical appraisals.

Study	Region	ML/DL models	Hydrological models	Key findings	Critical appraisal
[33]	Global review – surface water management	Review of LSTM, CNN, GNN, Transformers, and RL	SWAT, HEC-HMS, WEAP	ML/DL widely outperforms in specific tasks (forecasting, QA), but needs governance and XAI.	Strength: systems perspective; Limitation: high-level, few basin-scale comparisons.
[30]	Tarim River glacial basins	GRU; hybrid GRU-GSWAT+	GSWAT+ (SWAT+)	Hybrid reduced bias, improved low/high flow capture versus GSWAT+ and standalone GRU.	Strength: preserves physics, reduces bias; Limitation: data paucity for validation, opaque residual learning.
[54]	Synthetic experiments / hillslope-link model	GRU-Attention (DL)	Distributed hillslope-link hydrological Model (HLM)	DL can identify rainfall-runoff mapping in hindcast mode; however, there is less advantage in forecast mode, where future rainfall is uncertain.	Strength: controlled diagnostics; Limitation: synthetic → real-world complexities.
[88]	Indian tropical basins (future projection)	LSTM (DL)	Benchmark ML models	LSTM outperformed ML benchmarks for climate-sensitivity streamflow projections (high NSE and correlation).	Strength: multi-scenario projections; Limitation: relies on bias-corrected GCM forcing.
[90]	Sava River Basin (precipitation reconstruction)	ML/DL (various)	Traditional regression / dendro proxies	ML/DL improved precipitation reconstructions compared to linear regression.	Strength: improved paleo reconstructions; Limitation: extreme magnitude differences with streamflow proxies.
[94]	Sutlej (Himalaya) glacierized catchment	SVM, RF, DNN, BLSTM	No single conventional comparison to gauge records	BLSTM best overall; ML models performed well using EO & reanalysis in data-poor basin.	Strength: uses EO data; Limitation: short gauge record and transferability uncertain.
[52]	Wadi Mina (Algeria) sediment transport	DNN, CNN, GRU, Cat Boost	Sediment rating curves (SR)	DNN outperformed SRC and other ML models (NSE ≈ 0.99).	Strength: substantial quantitative gains for sediment; Limitation: overfitting, region-specific.
[56]	Upper Indus Basin (glacio-hydrology)	CNN-LSTM hybrids with GSM-SOCONT outputs	GSM-SOCONT (glacio-hydrological model)	The hybrid CNN-LSTM (with glacier features) outperformed GSM-SOCONT in validation (NSE and KGE gains).	Strength: integrates process outputs; Limitation: interpretability of learned corrections.
[91]	Lake Abaya-Chamo (Ethiopia)	LSTM, RF, SVM for gap-filling	Spatial interpolation (IDW)	LSTM is superior to daily rainfall infilling; ML reduces errors compared to SI.	Strength: practical gap-filling; Limitation: preprint, not peer reviewed.
[51]	Review – ensemble ML in water resources	EML: Bagging, Boosting, Stacking	Survey across many models	Ensemble approaches grew in popularity after 2018 and often yield better generalization than single models.	Strength: systematic bibliometrics; Limitation: ensemble complexity and interpretability.

are imperfect, evaluation metrics (especially regarding extremes and low flows) are still being standardized - all of which argues that the ML/DL seems to complement rather than replace well-parameterized hydrological models where decision-grade, mechanistic understanding is required.

Table 4 shows that physical structure, constrained learning in hybrid and physics-informed models, always reduces systematic bias and increases robustness across hydrological regimes [30,56,95]. However, standalone DL models may be more efficient than hybrid models used in data-rich hindcast-based applications and for small-scale predictions and short lead times [45,88,89]. Ensemble ML models can be more accurate, stable, and interpretable, especially when using tabular hydroclimatic data [96,97].

4.2. Common algorithmic structures and data types for river applications

The increased proportion of hydrological forecasting research on water quality and geomorphology in the present review suggests an unequal representation in the current literature, rather than a selection bias. The ability to use long-term gauge data and to create and standardize evaluation metrics is advantageous for streamflow and flood forecasting and helps develop and benchmark ML/DL models [4,5,33,65]. Conversely, the processes of sediment transport, river morphology, and water quality are usually observed on an ad-hoc basis, are controlled by site-specific factors, and are associated with measurement error, reducing the magnitude and comparability of ML/DL applications [52,98]. However, the representative research shows that, when physical constraints or uncertainty-conscious learning are considered, DL and hybrid models can significantly enhance predictions of sediment yield, channel dynamics, and water quality indices [5,52,80,99], and these areas are the key opportunities for methodological expansion in the future.

Supervised learning algorithms and hybrid metaheuristic models are primarily used in ML/DL for river studies [16,96,100]. The algorithms are mainly determined by data dimensionality, the non-linearity of the hydrological process, and spatial-temporal scales [101].

Traditional neural networks (MLP, ANN) and SVM algorithms have become the standard for regression-type hydrological predictions due to their ability to model non-linear input-output relationships [98]. However, they are constrained by local minima problems and a reliance on hyperparameter optimization [102]. Elshorbagy et al. (2010) indicated that SVMs and genetic programming (GP) produce more stable and precise structures than ANNs in rainfall-runoff modeling, with SVMs exhibiting faster convergence and better generalization on sparse data [100]. On the other hand, ANNs are more effective for continuous discharge predictions and evapotranspiration that require sufficient temporal information, but this comes at the cost of interpretability [65,102].

The predictive accuracy of DL algorithms, especially LSTM and CNNs, is higher because they can extract sequential and spatial features [71,103]. A hybrid CNN-BAT outperformed ANFIS-BAT, SVM-BAT, and RF-BAT models in daily streamflow prediction due to the ability of CNN to predict non-stationary rainfall-runoff interactions with rainfall autocorrelation [103]. Similarly, Cat Boost and XGBoost outperformed the standalone LSTM and regression models in predicting daily river discharge in the Sakarya Basin of Turkey [96] because the genetic algorithm optimized hyperparameters and avoided overfitting.

GeoAI that combines ML/DL with geospatial data using clustering, regression, and optimization [104]. It performed better in spatial mapping and classification (sediment yield and fluvial geomorphology) using RF and CNNs, whereas calibration and multi-objective optimization are better performed using evolutionary algorithms [104]. Hybrid algorithms have different performance depending on the type of data [19]. RFs and boosting trees perform better when the data is structured

as tabular or categorical, with medium feature interdependence [99], but CNNs and LSTMs perform better with dense spatiotemporal sequences [19].

Data quality and representativeness are the most critical issues in ML/DL based modeling [105]. DL models achieve high predictive accuracy when sensor networks are dense, but degrade quickly when they are sparse [16]. Hybrid metaheuristic-DL models reduce it by improving feature optimization, but require substantial computational resources and remain limited to physical interpretation [106]. Therefore, it is recommended to transition to hybrid PINNs to enhance model transferability and better understand the processes [94].

Therefore, predictive performance across a range of river tasks is rated based on the model's complexity and the data's heterogeneity, using a ranking system that prioritizes hybrid DL over DL, ensemble ML, single ML models, and statistical baselines.

4.3. Use of hybrid and physics-informed ML/DL systems for understanding hydrological processes

Hybrid, physics-informed ML/DL models establish a new frontier in river science [33,112]. For example, Xu et al. (2024) showed that combining DL with physically based hydrological models enhanced monthly streamflow forecasts by maintaining mass balance and other hydrological principles, which are frequently overlooked in data-driven-only models [95]. Similarly, Yifru et al. (2024) used a hybrid DL model that combines watershed memory and process-based attributes, thereby improving the capacity to simulate non-linear flow processes and to generalize across different hydroclimatic conditions [113]. This hybridization can help address the black-box problem often associated with DL models by incorporating physics-based priors that guide learning toward physically realistic solutions.

These hybrid models have helped amplify the mechanistic understanding of hydrological processes in recent years. A physics-informed, data-driven model of irrigated watersheds that constrained network learning using the differential equations of water balance achieved superior process fidelity and understandability [114]. Similarly, Bhasme et al. (2022) developed a physics-informed ML that was more effective at predicting streamflow and evapotranspiration and was physically consistent between inputs and outputs, and demonstrated that the

addition of governing equations to neural structures could constrain model behavior within hydrological realism [115].

The hybrid and physics-informed ML/DL models should be more successful than purely data-driven ones because they are confined to the space of physically plausible solutions, thereby reducing equifinality and improving extrapolation when seeking non-stationary solutions [26, 116]. These models also incorporate mass balance, routing, or energy conservation concepts to eliminate physically inconsistent predictions that can arise from purely statistical optimization [95,115,117]. However, these benefits come at the cost of more complex models, higher computational costs, and even the inability to determine whether performance gains are due to physical bottlenecks or to the model's ability to learn flexible residuals [20,56,92].

The problem of interpretability is critical: Slater et al. (2023) believe that the proliferation of ML models in hydrology will necessitate the implementation of XAI systems that will decompose the influence of the physical drivers and deliver understandable decisions [20].

Furthermore, Table 5 summarizes current developments in hybrid and physics-informed ML/DL models for river studies. These methods integrate both data-driven and physically based techniques to improve predictive accuracy, generalization, and interpretability. This study found that streamflow simulations are consistently improved, overfitting is reduced, and physical consistency is enhanced compared to traditional standalone ML models.

4.4. Key technical and ethical issues restricting ML/DL practical implementation in river basin management

4.4.1. Data access, completeness, and representativeness

Heterogeneous data access and quality are the basic constraints of the practical implementation of ML/DL in river basins [105]. The volume and heterogeneity are enhanced by remote sensing, in situ sensors, and UAV observations; however, these data can be spatially or temporally incomplete, thereby training the model to learn towards observed regimes [11,21]. The basins with limited data (developing regions) have limited gauges and telemetry, which compromise model training and transferability [54,119]. Additionally, the use of bias-corrected GCM forcing to project scenarios can cause non-stationary error structures to propagate into ML forecasts [88]. The gaps are addressed by

Table 5
Critical comparative summary of algorithmic structures and data types in river applications.

Algorithmic structure	Data type(s) used	Application	Performance and remarks	Reference
XGBoost, CART, RF (tree-based ML models)	Climatic (temperature, precipitation), hydrological (streamflow) data from Upper Indus, Chenab, Jhelum, and Kabul catchments	Streamflow simulation in data-scarce, glacier-fed, high-elevation catchments	R^2 and NSE between 0.61–0.90; SHAP identified minimum temperature as the key predictor; strong interpretability for hydrologic relevance	[97]
ANN, SVM, ANFIS	Precipitation (P), Temperature (T), Groundwater level (G), Lake level (L)	Groundwater level prediction considering surface water–groundwater interaction (Lake Okeechobee, Florida)	Including lake level (L) improved accuracy; ANFIS and SVM outperformed ANN; dependable for 1–3-month forecasts	[107]
Ensemble ML (AdaBoost + DT)	Elevation, Slope, Aspect, Rainfall, Land use, NDVI, Soil Hydrology, Proximity to River, Distance from Landslides, TWI, SPI, STI	Flood susceptibility mapping in Haraz-Neka River Basin (Mazandaran Province)	The ensemble model outperformed individual ML models: AUC > 0.96, accuracy 0.93, sensitivity 0.95, specificity 0.92; enhancing flood-risk identification and mitigation planning.	[108]
ANFIS and Genetic Algorithm (GA)	Meteorological data (rainfall, temperature, evaporation)	Runoff estimation in Sivand River Basin, Iran	ANFIS outperformed GA, with lower RMSE and MAE and higher NSE and R^2 .	[109]
CNN (optimized with BAT)	Multivariate rainfall–streamflow sequences	Daily discharge prediction	$r \approx 0.95$; best among compared models; efficient after optimization	[103]
LSTM optimized	Rainfall and river water level data	River water level forecasting	Bayesian Search achieved the best accuracy and efficiency; model performance is sensitive to hyperparameters.	[110]
Cat Boost + GA	Daily discharge, climatic inputs	River flow forecasting	Outperformed LSTM and linear regression ($R^2 > 0.9$); efficient and stable	[96]
RF	Hydro-climatic data from 95 catchments (Canada, USA)	Flood event and peak discharge simulation (daily scale)	Competitive with the conceptual Model (Hydromad) for low–medium floods; less accurate for high floods; performance influenced by climate and elevation; low-cost alternative for large-scale simulations	[111]

Table 6
Hybrid and physics-informed ML/DL applications in river studies.

Study	Model Type	Target Variable	Integration Method	Key Findings
[95]	DL + Physically based model	Streamflow	Coupling with hydrological equations	Improved monthly flow prediction and physical consistency
[113]	Hybrid DL (Watershed memory + process model)	Streamflow	Process-based features	Enhanced non-linearity capture and generalization
[114]	Physics-informed NN	Hydrological processes	PDE constraints	Improved interpretability and physical consistency
[115]	Physics-Informed ML	Streamflow, ET	Hydrological law regularization	Reduced overfitting and improved realism
[118]	CNN-LSTM Hybrid	Streamflow	Temporal convolution + LSTM	Superior flood prediction accuracy
[99]	ML hybrid (ensemble)	Water Quality Index	Uncertainty modeling	Improved robustness under variability
[57]	Physics-AI Hybrid	Streamflow & flood	Neural integration in PDE solver	Enabled national-scale operational modeling
[20]	ML + XAI	Hydrological processes	Explainability frameworks	Highlighted interpretability challenge in hybrid models

data-synthesis methods (GANs, TimeGAN), but these methods may introduce synthetic biases when the extremes of variability (or individual events of compounds) are under-represented [38,78].

4.4.2. Model understandability and interpretability

Most highly performing architectures (LSTMs, transformers, CNNs) are black boxes, which is harmful to mechanistic understanding needed to make management decisions in which mass balance, routing, or legal responsibility are relevant [30,76]. A lack of drivers to model responses diminishes stakeholders' and regulators' trust [13]. It can be mitigated using explainable AI (XAI) techniques (feature importance, surrogate models, attention visualization), which do not always give physically consistent explanations [22,74]. Residual learning (also known as hybrid and PINNs) conserves the laws of physics and enhances interpretability, but it introduces model-design complexity and necessitates coupling strategies that are yet to be standardized [56,116].

4.4.3. Computational scalability and operational constraints

Ensemble systems and deep architectures require significant computational resources for both training and inference [19,51]. In operation centers with limited infrastructure, latency, and energy costs, these constraints limit real-time forecasting and early warning. The membrane/CPU requirements of distributed and semi-distributed hydrological models are also high when using high-resolution EO inputs or GNN/transformer structures. Surrogate modeling (emulators), model compression, and cloud/edge strategies are helpful but create governance and data privacy concerns.

4.4.4. Generalizability, transferability, and uncertainty quantification

A key limitation of ML/DL models in hydrology is their limited generalizability to basins with different hydroclimatic regimes and physical properties. Single-catchment-trained models tend to reproduce local statistical relationships, but not transferable hydrological processes, and therefore have poorer outcomes on ungauged or climatically different basins [30,94,95]. Varying LULC, topography, soil characteristics, and climate forcing change relationships between rainfall and

runoff and sediment dynamics, and thus break the assumption that training and prediction data are uniformly distributed [27,77]. These issues are only magnified in the case of climate non-stationarity, where hydrological responses change with time as a result of anthropogenic and environmental changes [76].

Several methodological approaches are suggested to enhance model transferability and robustness. Shared representations. Fine-tuning parameters or transfer learning and regionalization strategies allow models that have been trained in data-rich basins to be applied to ungauged or data-sparse basins [29,77]. Similarly, the meta-learning and multi-basin training systems demonstrated the ability to capture generalized hydrological patterns in a variety of hydroclimatic regimes and, therefore, enhance predictive performance beyond the training range [9]. In addition, hybrid and physics-informed ML models are also reported to be more transferable by constraining learning to physical laws such as the water balance or energy conservation, which reduces a model to local statistical correlations [95,116].

Another aspect of the operationalization of ML/DL models in river basin management is the uncertainty quantification. Hydrological forecasts are always uncertain because of input uncertainty (e.g., rain forecasts), parameter uncertainty, model structural uncertainty, and observational errors [6]. One of the most common methods to model predictive uncertainty is ensemble modeling. A method of combining a large number of models or a large number of realizations of a single model to obtain a distribution of predictions, rather than a deterministic one, is an ensemble method: it may be bagging, boosting, stacking, or a multi-model ensemble [51]. Ensemble ML algorithms like the RF or GB have the inherent benefit of providing uncertainty estimates due to variability between decision trees, which enhances model robustness and decreases variance [66,120]. Nonetheless, which is challenging to compute, ensemble complexity may make it difficult to assign uncertainty sources and affect the cost of computation.

Bayesian methods provide a probabilistic model of the explicit representation of parameter and predictive uncertainty. Bayesian neural networks, Bayesian model averaging, and Gaussian Process Regression approximate the posterior distribution of model parameters and outputs, using which a credible interval can be used in predicting results instead of a point estimate [60]. Bayesian inference has also been popularly applied in hydrological modeling to propagate uncertainty in the parameters and assess the structural reliability of the model [121]. Bayesian methods are costly in calculation, though they increase interpretability and provide statistically sufficient limits of uncertainty that can be employed in risk-based water management.

In addition to ensembles and Bayesian methods, there are other methods that have been developed to quantify uncertainty in ML/DL-based hydrology. Uncertainty in input variables, model parameters, or climate scenarios is more often propagated with the help of Monte Carlo simulation through repeated sampling of a probability distribution [122]. Equally, the Generalized Likelihood Uncertainty Estimation model compares the parameter sets and derives a realization of the behavioral models by likelihood threshold to allow a probabilistic representation of the hydrological forecasts [35]. Other techniques, such as DL applications, include Monte Carlo dropout, quantile regression, and probabilistic forecasting networks, which estimate predictive uncertainty and are computationally efficient [123]. These techniques enable ANNs to estimate Bayesian inference and produce prediction intervals of hydrological variables, such as discharge or water level [35].

Through such developments, the inconsistent application of uncertainty analysis in hydrological research using ML/DL has not been done consistently. Numerous studies pay less attention to rigorous uncertainty assessment and more attention to predictive accuracy measures (e.g., NSE or RMSE, but do not consider extreme events and low-flow conditions when prediction errors may have a critical impact on the management [35,51]. Uncertainty measurement does not have standardized measures, and this further limits inter-study comparability and adoption of operations.

4.4.5. Ethical, legal, and social considerations

Algorithmic bias, unequal access to data/compute, and lack of transparency can exacerbate inequities in water management [40,119]. Privacy concerns arise from high-resolution monitoring of land use and households. Accountability for automated decisions (reservoir releases, flood warnings) remains unclear regarding legal liability. Responsible deployment demands governance frameworks, open data practices, and stakeholder participation to ensure equity and legitimacy [40,124].

Overcoming these barriers requires integrated solutions: open, standardized datasets; hybrid XAI-aware models that enforce physical constraints; computationally efficient emulators for operational use; and governance mechanisms that address ethics, equity, and accountability [116,125].

A practical case of ethical risk arises in ML-based early warning systems, where training data are often concentrated in well-instrumented basins. Models deployed in data-scarce or marginalized regions may therefore underestimate flood risk, disproportionately exposing vulnerable communities to harm while providing a false sense of security [40,119]. Without transparent communication of uncertainty and stakeholder oversight, automated reservoir release or warning decisions can obscure accountability and exacerbate social inequities, underscoring the need for governance frameworks alongside technical innovation [124].

4.5. Futuristic research directions to facilitate explainable, transferable, and integrated AI systems

4.5.1. Physics-aware and hybrid architectures with rigorous constraints

Develop hybrid systems that are defined to follow conservation laws and routing (e.g., PINNs, residual learning with mass-balance constraints) to balance predictive power with mechanistic validity [26,116]. The research should focus on modular strategies of coupling (loose or tight), error propagation analysis, and systematic benchmarking against traditional models [30,56].

4.5.2. Synthetic data, generative models, and extreme event augmentation

Train controlled generative models (i.e., TimeGAN, physics-guided simulators) for complex and extreme events in realistic open systems with the goal of both training robustly and simultaneously producing protocols to assess synthetic realism and absence of any unintended bias [38,61,78]. Researchers should explore adversarial testing suites to identify the model's significant failure modes for risk management [132].

Notice to model compression, surrogate emulators, and edge/cloud orchestration to enable real-time execution in resource-limited agencies [85]. Research into energy-efficient architectures, adaptive-fidelity simulation, and heterogeneous compute pipelines will allow scaling operations to [127].

4.5.3. Integrated multi-component and multi-scale systems

River flow, sediment transport, morphology, and water quality predictions can be advanced by spatio-temporal GNN and multi-task learning to capture inter-component interactions [128]. Multi-scale architectures further enable a balance between model fidelity and computational tractability [129].

Established community norms, shared data (metadata and provenance), and standard verification processes that are based on extremes, transferability, and interpretability [130,131]. Open-science repositories will reduce domestic disparities in perceived competence [132].

4.5.4. Ethical, governance, and participatory research

Research must include the social sciences to co-design AI tools with stakeholders, build accountability of automated decisions, and develop legal frameworks of data use and algorithmic transparency [133]. The privacy-preserving modeling (federated learning) and equity measures

are to be further investigated [131].

Interdisciplinary advances are required to achieve explainable, transferable, and integrated AI to support sustainable river management: physics-aware ML, hydrology-specific XAI, transfer/meta-learning, synthetic-data protocols, computationally efficient implementation, and open science and stakeholder-engaging governance systems [39,71,116,125].

5. Conclusion and recommendations

This systematic review shows that current studies are transforming the field of river studies to reproduce complex, non-linear hydrological and geomorphological processes. Pattern recognition, data fusion, and predictive accuracy of ML/DL models are superior to those of traditional physically based models, especially for non-stationary and heterogeneous data, but the standalone capacity of ML/DL models remains limited.

The use of physical constraints in ML/DL, including hybrid and physics-informed models, changes basic assumptions, improving generalization and interpretability, and reducing structural and parametric uncertainty. Nevertheless, there are uncertainties, including a lack of data, computational inaccessibility in developing countries, and inconsistent measures for assessing extremes and low-flow events.

This review guides the deployment of ML/DL models to improve river-based modeling. To promote mechanistic realism and enhance the transferability of models across basins: (1) it should focus on co-evolving hybrid ML-hydrological frameworks; (2), explainable and physical AI methods should be developed to increase transparency, interpretability, and trust among stakeholders; (3), standardized protocols, open repositories, and benchmark datasets are designed to support reproducibility and equitable access; (4), transfer learning, meta-learning, and generative modeling can be used to overcome the issue of data sparsity in ungauged or under-monitored basins; (5) ML/DL-based water management software must have ethical and governance-related aspects, such as algorithm transparency, data privacy, and fair distribution.

ML and DL have the transformative potential of river-basin modeling and management, if their development is informed by physical interpretability, computational efficiency, and ethical responsibility. Their combination with hydrological models and open-science practices is a definitive move toward sustainable, intelligent, and inclusive management of river systems.

List of Abbreviations

AI	Artificial Intelligence
ML	Machine Learning
DL	Deep Learning
LULC	Land Use Land Cover
CNN	Convolutional Neural Network
RNN	Recurrent Neural Network
LSTM	Long Short-Term Memory
RF	Random Forest
SVM	Support Vector Machine
DNN	Deep Neural Network
PIML	Physics-Informed Machine Learning
SWAT	Soil and Water Assessment Tool
HEC-HMS	Hydrologic Engineering Center – Hydrologic Modeling System
MODFLOW	Modular Finite-Difference Groundwater Flow Model
HEC-RAS	Hydrologic Engineering Center – River Analysis System
GSM-SOCONT	Glacier and Snow Melt Hydrological Model (Socont)
SWAPS	Soil-Water-Atmosphere-Plant Model
HBV	Hydrologiska Byråns Vattenbalansavdelning (Hydrologic model)
CEEMDAN	Complete Ensemble Empirical Mode Decomposition with Adaptive Noise

XGBoostExtreme Gradient Boosting
 kNNk-Nearest Neighbors
 GPRGaussian Process Regression
 ANNArtificial Neural Network
 MLPMultilayer Perceptron
 GRUGated Recurrent Unit
 GNNGraph Neural Network
 GCNGraph Convolutional Network
 AEAutoencoder
 VAEVariational Autoencoder
 GANGenerative Adversarial Network
 PINNPhysics-Informed Neural Network
 XAIExplainable Artificial Intelligence
 PRISMAPreferred Reporting Items for Systematic Reviews and Meta-Analyses
 SLRSystematic Literature Review
 NSENash-Sutcliffe Efficiency
 KGEKling-Gupta Efficiency
 RMSERoot Mean Square Error
 MAEMean Absolute Error
 MAPEMean Absolute Percentage Error
 R²Coefficient of Determination
 DODissolved Oxygen
 BODBiochemical Oxygen Demand
 CODChemical Oxygen Demand
 WANNWavelet Artificial Neural Network
 WLSTMWavelet Long Short-Term Memory
 BiLSTMBidirectional Long Short-Term Memory
 CNN-LSTMHybrid Convolutional Neural Network – Long Short-Term Memory
 MPCModel Predictive Control
 RSRemote Sensing
 SVCDSatellite Video Change Detection
 SYSU-CDSun Yat-sen University Change Detection dataset
 LEVIR-CDLarge-scale Land-Cover Change Detection dataset
 FDLdetForward Dictionary Learning Detector
 DPDynamic Programming
 SDPStochastic Dynamic Programming
 WEAPWater Evaluation and Planning System
 QAQuality Assessment

Funding

No Funding was used in this research.

CRediT authorship contribution statement

Mohsin Nawaz: Writing – review & editing, Supervision, Project administration, Formal analysis. **Muhammad Waqas:** Writing – original draft, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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